

CAR / AUTOMOTIVE DIGITAL MARKETPLACE

TECHNICAL BLUEPRINT

React + Laravel + MySQL

₱450,000 • 12 Weeks

1. Blueprint Purpose

This document defines the technical and functional blueprint for the ₱450,000 CAR / Automotive Buy-and-Sell Marketplace. It is intended to guide the project manager, business analyst, UI/UX designer, frontend developers, Laravel developers, QA team and DevOps team through implementation.

The architecture is deliberately designed as a modular monolith so the MVP can be delivered within budget while preserving clear domain boundaries for future scaling.

2. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Laravel / PHP 8.2+
Database	MySQL 8
Authentication	Laravel Sanctum
Real-Time	WebSocket implementation compatible with Laravel
Cache	Redis (optional/when required)
Queue	Laravel Queue
Storage	AWS S3 or equivalent
Web Server	Nginx
OS	Ubuntu Linux
Version Control	GitHub
CI/CD	GitHub Actions
Future Mobile	React Native

3. High-Level Architecture

The system consists of a React web application, Laravel API/backend, MySQL database, real-time WebSocket layer, media storage, recommendation engine, and administrative interface.

- React Buyer/Seller/Dealer Web → HTTPS/REST → Laravel API
- React Admin → Secure Admin API → Laravel
- Laravel → MySQL for transactional and master data
- Laravel → Redis for cache, queues and real-time support where needed
- Laravel → S3/object storage for vehicle and parts media
- Laravel → WebSocket layer for real-time messaging
- Laravel → Recommendation service/API for MVP recommendations

4. User Roles

Role	Primary Responsibilities
Buyer	Search vehicles, view details, favorites, recommendations, inquiries, offers, appointments, chat, parts shopping
Seller	Create vehicle listings, upload media/360°, manage price, inquiries, offers, appointments and chat
Dealer	Manage dealership profile, inventory, staff, leads, appointments and analytics
Parts Seller	Manage automotive parts, stock, pricing and orders
Admin	Manage users, listings, vehicles, parts, orders, moderation, analytics, permissions and settings

5. Core Modules

- Authentication & Authorization
- User and Profile Management
- Buyer Marketplace
- Seller Management
- Dealer Management
- Vehicle Master Data
- Vehicle Listings
- Vehicle Media
- 360° Vehicle Viewer
- Favorites
- AI-Assisted Recommendations
- Inquiries
- Offers
- Appointments / Test Drives
- Real-Time Messaging
- Car Parts Marketplace
- Cart and Orders
- Notifications
- Analytics
- Administration

6. Database Blueprint

Core relationship model:

- users → buyer_profiles / seller_profiles / dealer_users
- dealers → dealer_users → users
- vehicle_brands → vehicle_models → vehicles → vehicle_listings
- vehicles → vehicle_images / vehicle_videos / vehicle_360_images
- users → favorites → vehicle_listings
- users → conversations → conversation_participants → messages
- vehicle_listings → inquiries → offers → appointments
- part_categories → parts → part_images
- users → carts → cart_items → parts
- users → orders → order_items → parts

7. Key Database Tables

Table	Important Fields
users	id, name, email, mobile, password, role, status, verified_at, timestamps
buyer_profiles	id, user_id, preferences, location, timestamps
seller_profiles	id, user_id, verification_status, business_name, timestamps
dealers	id, owner_id, name, address, contact, status, timestamps
vehicle_brands	id, name, status
vehicle_models	id, brand_id, name, status
vehicles	id, seller_id, dealer_id, brand_id, model_id, year, variant,

	mileage, transmission, fuel_type, body_type, engine, color, condition, description, location, status
vehicle_listings	id, vehicle_id, price, listing_status, featured, views_count, favorites_count, published_at
vehicle_images	id, vehicle_id, image_url, sort_order, is_primary
vehicle_360_images	id, vehicle_id, image_url, frame_number, sort_order
favorites	id, user_id, vehicle_listing_id
conversations	id, type, created_at
conversation_participants	id, conversation_id, user_id
messages	id, conversation_id, sender_id, message, message_type, attachment_url, read_at, created_at
inquiries	id, buyer_id, listing_id, message, status, created_at
offers	id, buyer_id, listing_id, amount, status, created_at
appointments	id, buyer_id, listing_id, appointment_date, type, status
part_categories	id, name, status
parts	id, seller_id, category_id, name, sku, description, price, stock, status
part_images	id, part_id, image_url
carts	id, buyer_id, status
cart_items	id, cart_id, part_id, quantity, price
orders	id, buyer_id, total_amount, status, payment_status, shipping_status, created_at
order_items	id, order_id, part_id, quantity, price, subtotal

8. REST API Blueprint

Domain	Example Endpoints
Auth	POST /api/v1/auth/register, POST /api/v1/auth/login, POST /api/v1/auth/logout, GET /api/v1/auth/me
Vehicles	GET /api/v1/vehicles, GET /api/v1/vehicles/{id}, POST /api/v1/vehicles, PUT /api/v1/vehicles/{id}
Listings	GET /api/v1/listings, POST /api/v1/listings, PUT /api/v1/listings/{id}, DELETE /api/v1/listings/{id}
Favorites	GET /api/v1/favorites, POST /api/v1/favorites, DELETE /api/v1/favorites/{id}
Inquiries	GET /api/v1/inquiries, POST /api/v1/inquiries, GET /api/v1/inquiries/{id}
Offers	GET /api/v1/offers, POST /api/v1/offers, PUT /api/v1/offers/{id}
Appointments	GET /api/v1/appointments, POST /api/v1/appointments, PUT /api/v1/appointments/{id}
Parts	GET /api/v1/parts, GET /api/v1/parts/{id}, POST /api/v1/parts, PUT /api/v1/parts/{id}
Orders	GET /api/v1/orders, POST /api/v1/orders, GET /api/v1/orders/{id}, PUT /api/v1/orders/{id}
Chat	GET /api/v1/conversations, POST /api/v1/conversations, GET /api/v1/conversations/{id}/messages, POST /api/v1/conversations/{id}/messages

9. 360° Vehicle Viewer

- Seller uploads a sequence of vehicle images.
- Laravel validates and stores media.
- React loads the image sequence.
- Buyer drags/swipes to rotate the vehicle.
- Viewer supports zoom and responsive mobile display.

MVP uses image-sequence rotation. Full 3D modeling, photogrammetry and automated 360° capture are future features.

10. AI Recommendation Blueprint

- Capture searches, views, favorites and inquiries.
- Build buyer preference signals for brand, model, budget, category and location.
- Calculate a recommendation score.
- Return similar and personalized vehicle listings.
- Display 'Recommended for You', 'Similar Vehicles' and 'Within Your Budget'.

The MVP does not include training a proprietary machine-learning model. External AI/API usage, if selected, is subject to provider costs.

11. Real-Time WebSocket Chat

- Create conversation between buyer and seller/dealer.
- Authenticate WebSocket connection.
- Send message event.
- Persist message in MySQL.
- Broadcast message to recipient.
- Update read/unread state.
- Support basic image attachments.
- Trigger notification when appropriate.

Recommended data flow: React → WebSocket → Laravel real-time layer → Redis where required → MySQL persistence → recipient React client.

12. Car Parts Marketplace

- Parts categories and brands
- Product catalog
- Product details and media
- Search
- Stock and price
- Cart
- Basic checkout/order request
- Buyer order history
- Parts seller product/inventory management
- Admin product/order management

13. Analytics Blueprint

- Registered users by role
- Active listings
- Vehicle views
- Favorites
- Inquiries
- Offers
- Appointments
- Popular brands/models
- Listing performance

- Parts products
- Parts orders
- Basic sales/revenue summaries
- Date-range reporting

14. React Project Structure

- src/components/
- src/pages/Home/
- src/pages/Vehicles/
- src/pages/VehicleDetails/
- src/pages/Favorites/
- src/pages/Messages/
- src/pages/Appointments/
- src/pages/Parts/
- src/pages/Profile/
- src/admin/Dashboard/
- src/admin/Users/
- src/admin/Vehicles/
- src/admin/Sellers/
- src/admin/Dealers/
- src/admin/Parts/
- src/admin/Orders/
- src/admin/Analytics/
- src/services/api.js
- src/hooks/
- src/contexts/
- src/layouts/
- src/routes/
- src/utils/

15. Laravel Project Structure

- app/Http/Controllers/Auth/
- app/Http/Controllers/Buyer/
- app/Http/Controllers/Seller/
- app/Http/Controllers/Dealer/
- app/Http/Controllers/Vehicle/
- app/Http/Controllers/Parts/
- app/Http/Controllers/Order/
- app/Http/Controllers/Chat/
- app/Http/Controllers/Admin/
- app/Http/Requests/
- app/Http/Resources/
- app/Models/
- app/Services/
- app/Events/

- app/Listeners/
- app/Jobs/
- app/Notifications/
- app/Policies/

16. Recommended Service Classes

- VehicleService.php
- ListingService.php
- MediaService.php
- RecommendationService.php
- ChatService.php
- PartsService.php
- CartService.php
- OrderService.php
- AppointmentService.php
- AnalyticsService.php
- NotificationService.php

17. Authentication & Authorization

- Laravel Sanctum for API authentication
- Role-based authorization: Buyer, Seller, Dealer, Parts Seller, Admin
- Policies for resource-level permissions
- Secure password hashing
- Email/mobile verification as required
- Rate limiting on login and sensitive endpoints
- Admin audit logging

18. Deployment Blueprint

- Local Development → GitHub → Staging → QA/UAT → Production
- Production Nginx serves React assets and routes API traffic to Laravel.
- MySQL stores transactional data.
- Object storage stores vehicle/parts media.
- Redis supports cache, queue and real-time workloads when enabled.
- HTTPS is mandatory for production.
- Automated backups should be configured before launch.

19. Development Environments

Environment	Purpose
Local	Developer implementation and unit testing
Staging	Integration testing, QA and client UAT
Production	Live customer-facing platform

20. Git & JIRA Workflow

Recommended workflow:

JIRA Epic → Story → Task → Git feature branch → Pull Request → Code Review → QA → UAT → Production

- feature/vehicle-listing
- feature/360-viewer
- feature/ai-recommendation
- feature/parts-marketplace
- feature/realtime-chat
- feature/analytics

21. JIRA Epics

- EPIC 1 — Project Foundation
- EPIC 2 — Authentication & User Management
- EPIC 3 — Vehicle Marketplace
- EPIC 4 — Seller & Dealer Management
- EPIC 5 — 360° Vehicle Viewer
- EPIC 6 — AI Recommendations
- EPIC 7 — Real-Time Chat
- EPIC 8 — Car Parts Marketplace
- EPIC 9 — Analytics
- EPIC 10 — Administration
- EPIC 11 — QA, Security & Deployment

22. Development Sequence

- Project foundation and repositories
- Database schema and migrations
- Authentication and role management
- Vehicle master data
- Seller/dealer management
- Vehicle listings and media
- Buyer marketplace
- Favorites and inquiries
- Appointments
- 360° viewer
- Real-time chat
- Parts marketplace
- AI recommendations
- Analytics
- Admin completion
- QA, UAT and production launch

23. Security Blueprint

- HTTPS/TLS
- Laravel Sanctum authentication
- Authorization policies
- Request validation
- SQL injection protection through Eloquent/query bindings
- XSS protection
- CSRF protection where applicable
- API rate limiting
- Secure file upload validation
- Audit logs
- Secrets stored outside source code
- Database backup
- Access logging

24. Testing Strategy

Test Level	Coverage
Unit Testing	Services, validation, business rules
API Testing	Authentication, CRUD, authorization, error handling
Integration Testing	Database, storage, WebSocket, external services
UI Testing	Buyer, seller, dealer and admin workflows
Responsive Testing	Desktop, tablet and mobile browsers
Security Testing	Authentication, authorization, input/file validation
UAT	End-to-end business acceptance before launch

25. MVP Cost Allocation

Component	Budget
UI/UX Design	₱30,000
React Web Application	₱65,000
Laravel Backend/API	₱85,000
MySQL Database	₱25,000
Admin Panel	₱45,000
Buyer & Seller Modules	₱35,000
360° Vehicle Viewer	₱20,000
AI Recommendations	₱20,000
Car Parts Marketplace	₱45,000
Advanced Analytics	₱20,000
Real-Time WebSocket Chat	₱25,000
QA, Deployment & Security	₱20,000
Project Management	₱15,000
TOTAL	₱450,000

26. Timeline

Weeks	Deliverables
1–2	Requirements, UI/UX, architecture, database and project setup
3–5	Laravel backend, authentication, users, vehicle master data and listings
6–8	React marketplace, seller/dealer modules, search/filter and vehicle details

9–10	Parts marketplace, 360° viewer, recommendation engine and analytics
11–12	WebSocket chat, notifications, integration and security
13–14	QA, UAT and bug fixing, Production deployment, documentation and handover

27. MVP Limitations & Exclusions

- React Native dedicated mobile applications
- VIN scanner and paid VIN lookup
- Vehicle history provider integrations
- Custom AI/ML model training
- AI damage detection
- Full 3D vehicle modeling/photogrammetry
- Financing/bank integrations
- Insurance integrations
- Escrow services
- Advanced inspection network
- Large-scale dealer DMS/API synchronization
- Enterprise data warehouse / BI platform
- Multi-region high-availability infrastructure

28. Third-Party Costs

- AWS/cloud hosting
- Payment gateway fees
- SMS/OTP fees
- Email delivery
- Maps/geolocation APIs
- AI/API usage
- Vehicle data/history licensing
- App Store / Google Play fees
- External software licenses

29. Future Scaling Blueprint

Once the MVP has validated the business model, individual domains can be extracted into services as traffic and operational complexity justify it:

- User Service
- Vehicle Service
- Marketplace/Parts Service
- Chat Service
- Recommendation Service
- Order Service
- Notification Service
- Analytics/Data Service

Do not begin the ₱450,000 MVP as a large microservices system. A modular Laravel monolith is recommended for cost and delivery efficiency.

30. Definition of Done

- Feature implemented according to approved requirements
- Code committed to GitHub
- Code reviewed
- Unit/API tests completed where applicable
- QA passed
- Authorization verified
- Responsive UI verified
- UAT accepted
- Documentation updated
- Production deployment completed

31. Final Target

The target is a functional automotive marketplace that enables buyers to discover vehicles and parts, sellers/dealers to manage listings and leads, users to communicate in real time, and administrators to operate and analyze the platform.